0371. Sum of Two Integers

um | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/sum-c

Problem

- Add two integers without using + or -.

Intuition

- A ^ b adds the bits without carries, and (a & b) << 1 is exactly the carry.
- Repeat with (sum, carry) until the carry is 0. Two's complement makes negatives work for free with Java's 32-bit ints.

Brute Force

- Increment in a loop: $O(|b|)$, and it still uses ++.

Optimal Approach

- While b != 0: carry = (a & b) << 1; a = a ^ b; b = carry.
- Return a.

Worked Example

```
a = 2 (10), b = 3 (11).

xor=01  carry=100   a=1 b=4
xor=101 carry=0     a=5 b=0
answer 5
```

Java Solution

```java
public class SumOfTwoIntegers {
    public int getSum(int a, int b) {
        while (b != 0) {
            int carry = (a & b) << 1;
            a ^= b;
            b = carry;
        }
        return a;
    }
}
```

Complexity

- Time: O(32). Space: O(1).

Pitfalls

- In Python you must mask to 32 bits; Java ints wrap naturally.